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| 标题 | Cycloadditions of 4-Alkenyl-2-aminothiazoles with Nitroalkenes in the Formal Synthesis of Pramipexole: An Experimental and Computational Study |
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## 摘要（原文节选）

4-Alkenyl-2-dialkylaminothiazoles act as in – out dienes in [4 + 2] cycloaddition reactions with nitroalkenes, furnishing 2-amino-6-nitro-4,5,6,7-tetrahydrobenzothiazoles in moderate to good yields, accompanied by a subsequent 1,3-H migration. These transformations proceed with exquisite site-, regio-, and diastereoselectivity. This strategy is further enriched by revealing a novel route for pramipexole synthesis. The examination of the potential energy surfaces associated with the four possible reaction pathways for ⋯

## 数据表 · Table 2（16 行，含数值行 8 行）

ωB97X-D/6-31+G** Relative Gibbs Free Energies in kcal · mol – 1, Computed at 298 K in Toluene, of the Stationary Points Involved in the Diels – Alder Reaction of 4-Alkenyl-2-dimethylaminothiazoles 1b,e with Nitroethylenes 2a,b

| path | | 1b+2a | 1b+2b | 1e+2a | 1e+2b |
|---|---|---|---|---|---|
| endo-A | TS1xy-endoA | 26.6 | 23.6 | 24.7 | 22.0 |
| INTxy-endoA | 10.8 | 5.4 | | | |
| TS2xy-endoA | 22.5 | 17.5 | | | |
| 3 ′ xy-endoA | – 18.1 | – 22.6 | – 11.4 | – 16.1 | |
| 3xy-endoA | – 30.8 | – 34.7 | – 24.1 | – 28.6 | |
| exo-A | TS1xy-exoA | 23.7 | 22.5 | 20.3 | 20.7 |
| 9xy-exoA | 21.1 | | 16.4 | | |
| TS2xy-exoA | 21.8 | | a | | |
| 3 ′ xy-exoA | – 18.3 | – 24.8 | – 13.1 | – 19.8 | |
| 3xy-exoA | – 31.0 | – 36.2 | – 26.8 | – 31.2 | |
| endo-B | TS1xy-endoB | 29.9 | 25.7 | 29.6 | 27.2 |
| 3 ′ xy-endoB | – 14.9 | – 20.0 | – 11.4 | – 13.8 | |
| 3xy-endoB | – 25.1 | – 34.5 | – 25.4 | – 30.0 | |
| exo-B | TS1xy-exoB | 28.3 | 25.5 | 28.8 | 27.2 |
| 3 ′ xy-exoB | – 17.0 | – 22.9 | – 12.6 | – 18.1 | |
| 3xy-exoB | – 29.1 | – 35.7 | – 25.5 | – 31.5 | |

脚注（逐字）：aWe have not been able to optimize this transition state.

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